1/4/2026

Recall:

Finding PI if RHS is x^n

given eqn is:

$$f(D)y = x^n$$

Then $y_p = \frac{1}{f(D)} x^n$

$$= \frac{1}{1+Q(D)} x^n.$$

Eg 3: Solve $(D^2-4D+4)y = x^2+e^x+\cos 2x$

Soln: Aux. eqn is

$$D^2-4D+4=0$$

$\Rightarrow D = 2,2 \Rightarrow$ C.F is:

$$y_c = (c_1+c_2x)e^{2x}$$

P.I:

$$f(D) = D^2-4D+4$$

$$= 4\left(\frac{D^2}{4}-\frac{4D}{4}+1\right)$$

$$\Rightarrow f(D) = 4\left(1+\left(\frac{D^2}{4}-D\right)\right)$$

$\therefore$ PI is: (term by term)

$$y_p = \frac{1}{(D^2-4D+4)}(x^2+e^x+\cos 2x)$$

Now

$$y_{p_1} = \frac{1}{(D^2-4D+4)} x^2$$

$$= \frac{1}{4\left(1+\left(\frac{D^2}{4}-D\right)\right)} x^2$$

$$= \frac{1}{4}\left[1+\underbrace{\left(\frac{D^2}{4}-D\right)}_{\downarrow x}\right]^{-1} x^2$$

$$\left[(1+x)^{-1} = 1-x+x^2-x^3+\cdots\right.$$

$$\Rightarrow y_{p_1} = \frac{1}{4}\left[1-\overbrace{\left[\frac{D^2}{4}-D\right]}^{\nearrow x} + \left[\frac{D^2}{4}-D\right]^2\right]x^2$$

$$\Rightarrow y_{p_1} = \frac{1}{4}\left[1(x^2)-\frac{D^2}{4}(x^2) + D(x^2) + \left[\frac{D^4}{16}x^2-\frac{2D^3}{4}x^2+D^2x^2\right]\right]$$

$$\boxed{\begin{aligned} D(x^2) &= 2x \\ D^2(x^2) &= 2 \\ D^n(x^2) &= 0,\ n>2 \end{aligned}}$$

$$\Rightarrow y_{p_1} = \frac{1}{4}\left[x^2-\frac{1}{4}(2)+2x + 0-0+2\right]$$

$$\Rightarrow y_{p_1} = \frac{1}{4}\left(x^2+\frac{3}{2}+2x\right)$$

Now $y_{p_2} = \dfrac{1}{D^2-4D+4} e^x$.

$f(D) = D^2-4D+4$

Here $a=1$,

$f(1) = 1^2-4+4$

$= 1 \neq 0$

$\therefore y_{p_2} = \dfrac{1}{f(1)} e^x$

$= \dfrac{1}{1} e^x$

ie $y_{p_2} = e^x$,

$y_{p_3} = \dfrac{1}{D^2-4D+4} \cos 2x$

$a=2$, Replace D^2 by -2^2 in $f(D)$

$f(-2^2) = -\cancel{4} - 4D + \cancel{4}$

$= -4D \neq 0$

$\therefore y_{p_3} = -\dfrac{1}{4D} \cos 2x$

$= -\dfrac{1}{4} \int \cos 2x \, dx$

$\Rightarrow y_{p_3} = -\dfrac{1}{4} \times \dfrac{\sin 2x}{2}$

$\therefore y_p = y_{p_1} + y_{p_2} + y_{p_3}$

& the general solⁿ is: $y = y_c + y_p$

Type 4: RHS is $Q(x)$ where

$$Q(x) = e^{ax} \cdot P(x)$$

Where $P(x)$ is any function of x

eg $Q(x) = e^{5x} x^3 + e^{-3x} \cdot \cos(5x+7)$ $\rightarrow (*)$

We have $f(D)y = e^{ax} \cdot P(x)$

$$\Rightarrow y_p = \frac{1}{f(D)} e^{ax} \cdot P(x)$$

ie $y_p = e^{ax} \cdot \frac{1}{f(D+a)} P(x)$

For eg $(*)$, $y_p = \frac{1}{f(D)}\left(e^{5x} x^3 + e^{-3x} \cos(5x+7)\right)$

$$= e^{5x} \cdot \frac{1}{f(D+5)} x^3 + e^{-3x} \cdot \frac{1}{f(D-3)} \cos(5x+7)$$

Egs

1. Solve $(D^2 - 3D + 2)y = x^2 e^x$

Soln: Aux eqn is:

$$D^2 - 3D + 2 = 0$$

$$\Rightarrow D = 2, 1 \Rightarrow y_c = c_1 e^{2x} + c_2 e^x$$

Now **P.I**:

$$y_p = \frac{1}{(D^2 - 3D + 2)} x^2 \cdot e^x$$

$\boxed{a=1}$

$$= e^x \cdot \frac{1}{(D+1)^2 - 3(D+1) + 2} \cdot x^2$$

$$= e^x \cdot \frac{1}{D^2 + 2D + \cancel{1} - 3D - \cancel{3} + \cancel{2}}$$

$$\Rightarrow y_p = e^x \cdot \frac{1}{D^2 - D} x^2$$

$$\Rightarrow y_p = e^x \cdot \frac{1}{D^2 - D} x^2$$

$$= e^x \cdot \frac{1}{-D(-D+1)} x^2$$

$$\boxed{(1-x)^{-1} = 1 + x + x^2 + \cdots}$$

$$= e^x \cdot \left(\frac{-1}{D}\right)\left[1 - D\right]^{-1} x^2$$

$$= -e^x \cdot \frac{1}{D}\left[1 + D + D^2\right] x^2$$

$$= -e^x \cdot \frac{1}{D}\left[1(x^2) + D(x^2) + D^2(x^2)\right]$$

$$= -e^x \cdot \frac{1}{D}\left[x^2 + 2x + 2\right]$$

$$= -e^x \int (x^2 + 2x + 2)\, dx$$

$$\therefore y_p = -e^x \left\{ \frac{x^3}{3} + \frac{2x^2}{2} + 2x \right\}$$

& the general sol$^{\underline{n}}$ is:

$$y = y_c + y_p$$

2. Solve $(D^2 + 2)y = e^x \cos x + x^2 e^{3x}$

<u>Sol$^{\underline{n}}$</u>: Aux eq$^{\underline{n}}$ is:

$$D^2 + 2 = 0$$

$$\Rightarrow D = \pm \sqrt{2}\, i$$

$\therefore$ C.F is: $\underline{y_c = c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x}$

P.I :

$$y_p = \frac{1}{(D^2+2)}\left(e^x \cos x + x^2 e^{3x}\right)$$

$$= e^x \cdot \frac{1}{\left[(D+1)^2+2\right]} \cos x + e^{3x} \cdot \frac{1}{(D+3)^2+2} x^2$$

$a=1$ $\quad$ $a=3$

$$= e^x \cdot \frac{1}{D^2+2D+3} \cos x + e^{3x} \cdot \frac{1}{D^2+6D+11} x^2$$

$D^2 = -1^2$

$$= e^x \cdot \frac{1}{-1^2+2D+3} \cos x + e^{3x} \cdot \frac{1}{11\left(\frac{D^2}{11}+\frac{6D}{11}+1\right)} \cdot x^2$$

$$= e^x \cdot \frac{1}{2(1+D)} \cos x + \frac{e^{3x}}{11}\left[\left(1+\left(\frac{D^2}{11}+\frac{6D}{11}\right)\right)^{-1}\right] x^2$$

$\to x$

$$= e^x \cdot \frac{1-D}{2(1-D^2)} \cdot \cos x + \frac{e^{3x}}{11}\left[1-\left(\frac{D^2}{11}+\frac{6D}{11}\right)+\left(\frac{D^2}{11}+\frac{6D}{11}\right)^2\right] x^2$$

$$\Rightarrow y_p = \frac{(1-D)}{2(1-(-1^2))} \cos x + \frac{e^{3x}}{11}$$

$$\frac{e^x}{4}(\sin x + \cos x) + \frac{e^x}{11}\left[x^2 - \frac{12x}{11} + \frac{50}{121}\right]$$